6G4Z3006 Calculus: Mock exam 01

Duration : 3 hours

Instructions to students

- The 2025/26 exam will be a 3 hour exam.
- In the 2025/26 exam you will be given **two compulsory questions** with questions drawn from all parts of the module and then a choice of any **two** questions from a set of four optional questions (one question each based on the topics of sequences, series, differentiation and integration). Each of the four questions you will answer will be worth 25 marks.

Permitted materials

- Faculty formula handbook
- Students are permitted to use their own calculators without mobile communication facilities.

Section A – answer both questions

1. (a) Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. Use the concept of limiting value of a function to give the definition of the *derivative* of f at the point $a \in \mathbb{R}$. [3]

> **Solution:** The function f is differentiable at a and has derivative $f'(a)$ if the following limiting value exits,
>
> $$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$
>
> *3 for def*
> *accept $h \to 0$ version*

(b) Give an example of a continuous function $f : \mathbb{R} \to \mathbb{R}$ and a point $a \in \mathbb{R}$ in its domain such that f is *not* differentiable at a. Hint: you could define f as a certain piecewise-linear function. [3]

> **Solution:** For example, define $f : \mathbb{R} \to \mathbb{R}$ by
>
> $$f(x) = \begin{cases} -x & \text{, for } x < 0, \\ x & \text{, for } x \geq 0. \end{cases}$$
>
> This is continuous, but not differentiable at $a = 0$, as the limit in part (a) does not exist, (there are different left and right-handed limits as $x \to 0$).
>
> *3 for good example*
> *up to 2/3 if not continuous*

(c) State and prove the *sum rule* of differentiation. You can use without proof any relevant properties of the limiting values of functions, providing you clearly state these in your proof. [8]

> **Solution:** Consider the derivative $(f+g)'(a)$ of the sum of functions f and g. The main steps are
>
> $$\begin{aligned} (f+g)'(a) &= \lim_{x \to a} \left(\frac{(f+g)(x) - (f+g)(a)}{x - a} \right) \\ &= \lim_{x \to a} \left(\frac{f(x) + g(x) - \big(f(a) + g(a)\big)}{x - a} \right) \text{, def. of } f+g \\ &= \lim_{x \to a} \left(\frac{f(x) - f(a)}{x - a} + \frac{g(x) - g(a)}{x - a} \right) \text{, rewriting the fraction} \\ &= \lim_{x \to a} \left(\frac{f(x) - f(a))}{x - a} \right) + \lim_{x \to a} \left(\frac{g(x) - g(a))}{x - a} \right) \text{, linearity of limits} \\ &= f'(a) + g'(a), \text{ def. of derivs} \end{aligned}$$
>
> *8*
> *2 for each main step*

(d) State the *chain rule* of differentiation. [3]

> **Solution:** For functions f and g, differentiable at a and $f(a)$ respectively, the derivative of the composition $g \circ f$ is given by
>
> $$(g \circ f)'(a) = g'(f(a))f'(a).$$
>
> *3 for def.*

(e) State the *quotient rule* of differentiation. [3]

Solution: For functions a and v, differentiable at a and v non-zero at and about a, the derivative of the quotient function u/v is given by

$$(u/v)'(a) = \frac{v(a)u'(a) - u(a)v'(a)}{v^2(a)}.$$

3 for def.

(f) Give a proof of the quotient rule by considering the quotient u/v, of functions u and v, in the form [5]

$$u/v = u \cdot (r \circ v),$$

i.e. as the product of u with the composition r after v, where r is the reciprocal function defined by

$$r(x) = \frac{1}{x}.$$

You can use without proof the results of the product and chain rules, as well as the known derivative of the reciprocal function, i.e.

$$r'(x) = \frac{-1}{x^2}.$$

Solution: The main steps are

$$\begin{aligned}
(u/v)'(a) &= \big(u \cdot (r \circ v)\big)'(a) \\
&= u'(a)(r \circ v)(a) + (r \circ v)'(a)u(a), \text{prod. rule} \\
&= u'(a)/v(a) + r'(v(a))v'(a)u(a), \text{chain rule and def. of } r \\
&= u'(a)/v(a) - v'(a)u(a)/v^2(a), \text{deriv. of } r \\
&= \frac{v(a)u'(a) - u(a)v'(a)}{v^2(a)}, \text{tidy up.}
\end{aligned}$$

5
need to see main steps for full marks

Section A – answer both questions

2. (a) State both parts of the *Fundamental Theorem of Calculus*. Suppose the function $F(x)$ is defined by [8]

$$F(x) = \int_0^x (t^2 + \cos(t))dt.$$

Use the Fundamental Theorem of Calculus to find $F'(\pi/2)$, i.e. the derivative of F at $\pi/2$.

> **Solution: First part:** If f is a continuous integrable function and F is defined by $F(x) = \int_a^x f(t)\,dt$ then F is differentiable and $F'(x) = f(x)$.
>
> **Second part:** If f is continuous and integrable and F is an antiderivative of f, i.e. $F'(x) = f(x)$, then
>
> $$\int_a^b f(x)\,dx = F(b) - F(a).$$
>
> For the function F in the question we will get
>
> $$F'(\pi/2) = (\pi/2)^2 + \cos(\pi/2) = \pi^2/4.$$
>
> *3 for FTC 1.*
> *3 for FTC 2.*
> *2 for example*

(b) Suppose that $\{x_n\}_{n=1}^{\infty}$ is a sequence of real numbers. Give the (ϵ, N)-definition for the convergence of the sequence $\{x_n\}_{n=1}^{\infty}$. [3]

> **Solution:** The sequence $\{x_n\}$ converges to the limit L if and only if given any $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that if $n \geq N$ then $|x_n - L| < \epsilon$.
>
> *3 for the def.*

(c) Suppose that $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are two sequences of real numbers, and that $\alpha, \beta \in \mathbb{R}$. Use the definition from the previous part to prove that if $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are two convergent sequences with $a_n \to a$ and $b_n \to b$ as $n \to \infty$ then the linear combination sequence $\{\alpha a_n + \beta b_n\}_{n=1}^{\infty}$ is also convergent and [6]

$$\alpha a_n + \beta b_n \to \alpha a + \beta b, \text{ as } n \to \infty.$$

> **Solution:** Main steps in the argument are:
>
> - Choose $N_1, N_2 \in \mathbb{N}$ so that for all $n \geq N_1$ we have $|a_n - a| < \epsilon/(2|\alpha|)$ and for all $n \geq N_2$ we have $|b_n - b| < \epsilon/(2|\beta|)$
>
>   *2*
>
> - Set $N = \max(N_1, N_2)$. Then for all $n \geq N$ we have
>
> $$\begin{aligned} |(\alpha a_n + \beta b_n) - (\alpha a + \beta b)| &= |\alpha(a_n - a) + \beta(b_n - b)| \\ &\leq |\alpha||a_n - a| + |\beta||b_n - b|, \text{triang. ineq.} \\ &< |\alpha|\epsilon/(2|\alpha|) + |\beta|\epsilon/(2|\beta|) \\ &= \epsilon \end{aligned}$$
>
>   *4*
>
> - Hence $\alpha a_n + \beta b_n \to \alpha a + \beta b$, as $n \to \infty$.

(d) Use concepts of sequence convergence and partial sums to give the definition for the convergence of an infinite series $\sum_{n=1}^{\infty} a_n$. [3]

Solution: The partial sum, S_k, of the series is defined as

$$S_k = \sum_{n=1}^{k} a_n.$$

The series is said to converge to a sum L if and only if $S_k \to L$ as $k \to \infty$.

3

(e) Suppose that $\alpha, \beta \in \mathbb{R}$. Use parts (c) and (d) and relevant properties of sequence convergence to prove that if $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are two convergent series of real numbers with [5]

$$\sum_{n=1}^{\infty} a_n = a \text{ and } \sum_{n=1}^{\infty} b_n = b,$$

then the linear combination series $\sum_{n=1}^{\infty}(\alpha a_n + \beta b_n)$ is convergent also and

$$\sum_{n=1}^{\infty}(\alpha a_n + \beta b_n) = \alpha a + \beta b.$$

Solution: This involves working on the partial sum S_k of the new series and using the linear combinations rule to prove convergence. The main steps are

$$\begin{aligned} S_k &= \sum_{n=1}^{k}(\alpha a_n + \beta b_n) \\ &= \alpha\left(\sum_{n=1}^{k} a_n\right) + \beta\left(\sum_{n=1}^{k} b_n\right), \text{manip. of fin. sum} \\ &\to \alpha a + \beta b, \end{aligned}$$

where the last step follows from the linear combinations property of sequence convergence.

5
needs ref. to seq. conv. prop.
and clear manip. of fin. sum.
for full marks

End of Section A

Section B – answer any two questions

3. (a) Use the limit definition of the derivative to obtain the derivative of the function f defined by the formula [5]

$$f(x) = x^6.$$

Point out clearly where you make use of any properties of the limiting values of functions.

Solution: Implement the limit definition and evaluate as follows, for $a \in \mathbb{R}$,

$$\begin{aligned} f'(a) &= \lim_{h\to 0} \frac{(a+h)^6 - a^6}{h} \\ &= \lim_{h\to 0} \frac{a^6 + 6a^5h + 15a^4h^2 + 20a^3h^3 + 15a^2h^4 + 6ah^5 + h^6 - a^6}{h} \\ &= \lim_{h\to 0} (6a^5 + 15a^4h + 20a^3h^2 + 15a^2h^3 + 6ah^4 + h^5) \\ &= 6a^5, \end{aligned}$$

where the last step is justified by multiple applications of the linearity form of the algebra of limits theorem for limiting values of functions.

2 for correct set-up of limit
3 for completion
can also use $\lim_{x\to a}$ definition

(b) Consider the function g defined by the formula [6]

$$g(x) = x^4 \sin(2x).$$

Use the linearity, product and chain rules to obtain the first and second order derivatives, $g'(x)$ and $g''(x)$, of g. Point out clearly where you are using each of the rules.

Solution: Using the product rule and chain rules we get

$$g'(x) = 4x^3 \sin(2x) + 2x^4 \cos(2x).$$

Then using product and chain rules again we get, before simplification,

$$g''(x) = 12x^2 \sin(2x) + 8x^3 \cos(2x) + 8x^3 \cos(2x) - 4x^4 \sin(2x)$$

which tidies up to

$$g''(x) = (12x^2 - 4x^4) \sin(2x) + 16x^3 \cos(2x).$$

3 for $g'(x)$
3 for $g''(x)$

(c) Use the concepts of the *differrentiation* and *stationary point* to find the rectangle of maximal area that can be drawn inside the unit circle. [10]

Solution: Consider the unit circle in the plane $\mathbb{R}^2$, centred on the origin $(0, 0)$. A rectangle of maximal area drawn inside the circle will have all four corners on the circle itself at the four points $(\pm x, \pm y)$.

The rectangle will have width $2x$, height $2y$ and area A given by

$$A(x, y) = 4xy,$$

where x, y satisfy the circle constraint

$$1 = x^2 + y^2.$$

The circle constraint is equivalent to

$$y = \sqrt{1 - x^2},$$

and so the area can be written as

$$A(x) = 4x\sqrt{1 - x^2} = 4\sqrt{x^2 - x^4}.$$

The stationary points of A satisfy

$$\frac{dA}{dx} = 4\frac{1}{2}\left(x^2 - x^4\right)^{-1/2}\left(2x - 4x^3\right) = 0,$$

which is equivalent to

$$1 - 2x^2 = 0,$$

which has the solution $x = \pm 1/\sqrt{2}$.

So we have a single stationary point at $x = 1/\sqrt{2}$. The corresponding y value from the circle constraint is $y = 1/\sqrt{2}$ also.

4 for stationary point

That this is a local maximum can be justified by the fact that there is a single stationary point and clearly no minimum possible area as we can form inscribed rectangles with x arbitrarily close to zero, yet still with $y < 1$. Alternatively we can apply a second-order derivative test.

2 for maximum

So the rectangle with maximal area that can fit inside the unit circle is the square of side length $1/\sqrt{2}$.

1 for conclusion

(d) Let u, v, w, z be four differentiable functions of the variable x. Use a combination of the product rule and quotient rule to fully expand the derivative [4]

$$\frac{d}{dx}\left(\frac{uv}{wz}\right),$$

and show how it depends on the individual functions u, v, w, z and their derivatives.

Solution: Applying the quotient rule gives

$$\frac{d}{dx}\left(\frac{uv}{wz}\right) = \frac{wz\frac{d}{dx}(uv) - uv\frac{d}{dx}(wz)}{w^2z^2}.$$

1

Now apply the product rule a couple of times for the term in the numerator and simplify to get

$$\frac{d}{dx}\left(\frac{uv}{wz}\right) = \frac{u'vwz + uv'wz - uvw'z - uvwz'}{w^2z^2}$$

3

Section B – answer any two questions

4. (a) Evaluate the following integrals. You can use various integration techniques and/or trigonometric formula but you must explain what you are doing and clearly reference any standard integrals you make use of.

(i) [6]

$$I_1 = \int_0^{\pi/2} x^2 \sin(x)\, dx$$

Solution: This requires integration by parts twice.

$$\begin{aligned} I_1 &= \left[-x^2\cos(x)\right]_0^{\pi/2} - \int_0^{\pi/2} -\cos(x) 2x\, dx, \\ &= 0 + 2\int_0^{\pi/2} x\cos(x)\, dx, \\ &= 2\left([x\sin(x)]_0^{\pi/2} - \int_0^{\pi/2} \sin(x)\, dx \right), \\ &= 2\left(\frac{\pi}{2} + [\cos(x)]_0^{\pi/2}\right), \\ &= 2\left(\frac{\pi}{2} - 1\right), \\ &= \pi - 2. \end{aligned}$$

2 for each int. by parts and 2 for completion

(ii) Use the substitution $u = \sqrt{x+2}$ to evaluate the integral [6]

$$I_2 = \int_{-2}^{2} \frac{\sqrt{x+2}}{x+6}\, dx$$

Solution: By substitution, using $u = \sqrt{x+2}$, then $u^2 = x+2$, so $2u\, du = dx$. When $x = -2, 2$ then $u = 0, 2$.

$$\begin{aligned} I_2 &= \int_0^2 \frac{u}{u^2+4} 2u\, du, \\ &= \int_0^2 \frac{2u^2}{u^2+4}\, du, \\ &= \int_0^2 2 - \frac{8}{u^2+4}\, du \\ &= 4 - 4\int_0^2 \frac{2}{u^2+2^2}\, du \\ &= 4 - 4\left[\tan^{-1}\left(\frac{u}{2}\right)\right]_0^2 \\ &= 4 - 4\left[\frac{\pi}{4}\right] \\ &= 4 - \pi \end{aligned}$$

4 for transformation to u integral
2 for rest

(b) The Riemann integral of a function $f(x)$ between the limits $x = a$ and $x = b$ is defined as [13]

$$\int_a^b f(x)\,dx = \lim_{\substack{n\to\infty \\ \Delta x\to 0}} \left(\sum_{i=1}^{n} f(x_i^*)\Delta x_{i-1}, \right),$$

where the x_i $(0 \le i \le n)$ form a subdivision of the interval $[a, b]$ with

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b,$$

$\Delta x_{i-1} = x_i - x_{i-1}$ is the width of the i^{th} subinterval,
$\Delta x = \max(\{\Delta x_i \,:\, 0 \le i \le n-1\})$ and x_i^* is a representative point from the i^{th} subinterval, i.e. $x_{i-1} \le x_i^* \le x_i$.

In this question you will evaluate the Riemann integral for the function $f(x) = x^2$ and the limits $a = 3$ and $b = 5$. You should use the equally spaced interval

$$x_i = 3 + i\frac{2}{n}, \quad (i = 0, \dots, n),$$

and the start point of each subinterval as the representative point, i.e.

$$x_i^* = 3 + (i-1)\frac{2}{n}.$$

Set up and evaluate the limit in the Riemann integral, using known summation formulae, to show that

$$\int_3^5 x^2\,dx = 98/3.$$

You can use, without proof, any of the following summation formulae

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2},$$

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}.$$

Solution: Applying the Riemann integration formula to the given set up results in

$$I = \int_3^5 x^2\,dx = \lim_{n\to\infty} \left(\sum_{i=1}^{n} \left(3 + (i-1)\frac{2}{n}\right)^2 \frac{2}{n} \right)$$

3 for implmenting the Riemann integral formula

$$\begin{aligned} I &= \lim_{n\to\infty} \left(\sum_{i=1}^{n} \frac{2}{n} \left(9 + \frac{12}{n}(i-1) + \frac{4}{n^2}(i-1)^2 \right) \right) \\ &= \lim_{n\to\infty} \left(\sum_{i=1}^{n} \left(\frac{18}{n}\right) + \sum_{i=1}^{n} \left(\frac{24}{n^2}(i-1)\right) + \sum_{i=1}^{n} \left(\frac{8}{n^3}(i-1)^2\right) \right) \end{aligned}$$

3 for producing the three sums

$$\begin{aligned} I &= \lim_{n\to\infty} \left(\frac{18}{n} \sum_{i=1}^{n} (1) + \frac{24}{n^2} \sum_{i=1}^{n} (i-1) + \frac{8}{n^3} \sum_{i=1}^{n} (i-1)^2 \right) \\ &= \lim_{n\to\infty} \left(18 + \frac{24}{n^2} \sum_{i=1}^{n-1} i + \frac{8}{n^3} \sum_{i=1}^{n-1} i^2 \right) \end{aligned}$$

2 for simplifying the sums

$$I = \lim_{n\to\infty}\left(18 + \frac{24}{n^2}\frac{(n-1)n}{2} + \frac{8}{n^3}\frac{(n-1)n(2n-1)}{6}\right)$$

2 for implmenting the summation formulae

$$\begin{aligned} I &= \lim_{n\to\infty}\left(18 + 12\frac{1-1/n}{1} + \frac{4}{3}\frac{2-3/n+1/n^2}{1}\right) \\ &= 18 + 12 + 8/3 \\ &= 98/3 \end{aligned}$$

3 for taking the limit and reaching correct integral

Section B – answer any two questions

5. (a) Let the sequence $\{b_m\}_{m=2}^{\infty}$ be defined by the formula [5]

$$b_m = \frac{3m-5}{m}.$$

By considering the ratio or difference of its consecutive terms, prove that this sequence is increasing for all $m \geq 2$.

Solution: Examination the ratio and comparing it to 1 we get

$$\begin{aligned}\frac{b_{m+1}}{b_m} &= \frac{3m-2}{m+1}\frac{m}{3m-5}\\ &= \frac{3m^2-2m}{3m^2-2m-5}\\ &> 1, \text{ for all } m \geq 2,\end{aligned}$$

since the denominator is less than the numerator. Therefore $b_{m+1} > b_m$ for all $m \geq 2$ and the sequence is increasing.

2 for ratio formation
3 for rest

(b) Let the sequence $\{c_m\}_{m=1}^{\infty}$ be defined by the formula [7]

$$c_m = \frac{3m^3-2m+1}{-4m^3+2}.$$

Show how the algebra of limits theorem for convergent sequences can be used to prove that $\{c_m\}$ is a convergent sequence and to determine its limit. Make it clear in your answer where you are using each part of the theorem that you rely on.

Solution:

$$\begin{aligned}\lim_{m\to\infty} c_m &= \lim_{m\to\infty} \frac{3m^3-2m+1}{-4m^3+2}\\ &= \lim_{m\to\infty} \frac{3-2/m^2+1/m^3}{-4+2/m^3}\\ &= \frac{-3}{4},\end{aligned}$$

where the last step uses the quotient and linear combinations form of the algebra of limits theorem and the fact that $1/m^2, 1/m^3 \to 0$ as $m \to \infty$.

2 initial prep
2 quotient case
2 linear combs case
1 for rest

(c) When the Newton-Raphson method for root finding is applied to the equation

$$f(x) = x^2 - 5 = 0,$$

it produces a sequence, $\{x_n\}_{n=0}^{\infty}$, of rational estimates of $\sqrt{5}$, beginning with the number $x_0 = 3$ and for $n \geq 0$ the subsequent estimates are given by the recurrence relation

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2-5}{2x_n}.$$

In this question you will construct a proof that justifies the claim of the Newton-Raphson method, i.e. that the sequence does indeed converge to the limit $\sqrt{5}$.

(i) Use the recurrence relation above to prove, by induction, that for all $n \geq 0$ the sequence elements satisfy $x_n > \sqrt{5}$. [6]

> **Solution:** Base case is true as $x_0 = 3 > \sqrt{5}$. Assume that $x_n > \sqrt{5}$. Squaring the recurrence relation gives us
>
> $$\begin{aligned} x_{n+1}^2 &= \left(x_n - \frac{x_n^2 - 5}{2x_n}\right)^2, \\ &= 5 + \left(\frac{x_n^2 - 5}{2x_n}\right)^2, \\ &> 5, \text{ since, by assumption, } x_n^2 > 5 \text{ and } x_n > 0. \end{aligned}$$
>
> Hence, by induction, for all $n \geq 0$, $x_n > \sqrt{5}$.

(ii) Use the result of part (i) to prove that the sequence $\{x_n\}_{n=0}^{\infty}$ is a decreasing sequence. [2]

> **Solution:** To prove the decreasing nature of the sequence we compare $x_{n+1} - x_n$ to 0. From the recurrence relation we have, for all $n \geq 0$,
>
> $$x_{n+1} - x_n = -\frac{x_n^2 - 5}{2x_n} < 0,$$
>
> since, from part (i), $x_n^2 > 5$ and $x_n > 0$. So for all $n \geq 0$, $x_{n+1} < x_n$, i.e. the sequence is decreasing.

(iii) Briefly explain how the Monotone Convergence Theorem can be used to prove that $\{x_n\}_{n=0}^{\infty}$ is a convergent sequence. [2]

> **Solution:** From parts (i) and (ii), the sequence is decreasing and bounded below by $\sqrt{5}$. Therefore, by the Monotone Convergence Theorem, it is convergent (and has a limit $\geq \sqrt{5}$).

(iv) Finally, use the algebra of limits theorem to prove that $x_n \to \sqrt{5}$ as $n \to \infty$. [3]

> **Solution:** Let a denote the limit, i.e. $x_n \to a$ as $n \to \infty$. Taking the limit of both sides of the recurrence relation and applying the algebra of limits theorem, gives us
>
> $$a = a - \frac{a^2 - 5}{2a},$$
>
> which simplifies to
>
> $$a^2 = 5,$$
>
> so $a = \sqrt{5}$, since the negative root is not possible since we know from before that $a \geq \sqrt{5}$.

Section B – answer any two questions

6. (a) Consider the infinite series [8]

$$\sum_{n=1}^{\infty} \frac{2+3^n}{5^{n+2}}.$$

Use the result about the convergence of geometric series to prove that the series defined above is convergent and determine its sum. You will also need to make use of the theorem about linear combinations of convergent series.

Solution: Split the series up into two geometric series as follows and appeal to the geometric series result and theorem 2.2

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{2+3^n}{5^{n+2}} &= \sum_{n=1}^{\infty} \frac{2}{5^3}\left(\frac{1}{5}\right)^{n-1} + \sum_{n=1}^{\infty} \frac{3}{5^3}\left(\frac{3}{5}\right)^{n-1} \\ &= \frac{2/5^3}{1-1/5} + \frac{3/5^3}{1-3/5} \\ &= \frac{2}{25}\end{aligned}$$

8

(b) (i) Use a suitable comparison test to prove that the series [5]

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}}{2+n^2},$$

is a convergent series. You can make use of the known convergence of any hyper-harmonic series.

Solution: We compare to the convergent series $\sum 1/n^{3/2}$, using the limit comparison test. The limit to examine is

$$\begin{aligned}\lim_{n\to\infty}\left(\frac{\sqrt{n}}{2+n^2}\cdot n^{3/2}\right) &= \lim_{n\to\infty}\frac{n^2}{2+n^2} \\ &= \lim_{n\to\infty}\frac{1}{2/n^2+1} \\ &= 1, \text{ since } 2/n^2 \to 0 \text{ as } n\to\infty,\end{aligned}$$

Therefore by the limit comparison test and the fact that $\sum 1/n^{3/2}$ is a convergent hyper-harmonic series, the original series converges also.

2 correct comparator
3 execution and conclusion

(ii) Use the ratio test to prove that the series [5]

$$\sum_{m=1}^{\infty} \frac{3^m}{(m+2)!},$$

is a convergent series. *Note that* ! *denotes the factorial operation.*

Solution: Taking the limit of consecutive terms $x_m = \frac{3^m}{(m+2)!}$ we see

$$\begin{aligned}\lim_{m\to\infty} \frac{x_{m+1}}{x_m} &= \lim_{m\to\infty} \left(\frac{3^{m+1}}{(m+3)!} \frac{(m+2)!}{3^m} \right) \\ &= \lim_{m\to\infty} \left(\frac{3}{m+3} \right) \\ &= 0, \text{ by algebra of limits theorem.}\end{aligned}$$

Since the limit is 0, the original series converges by the ratio test as the limit has absolute value less than 1.

2 for correct start of of test
3 for limit of 0 and conclusion

(c) Obtain the partial fraction expansion of [7]

$$\frac{15}{m^2+5m+4},$$

and use it to obtain the sum of the infinite series

$$\sum_{m=1}^{\infty} \frac{15}{m^2+5m+4},$$

by considering cancellation effects in the partial sums of this series.

Solution: The required partial fraction expansion is

$$\frac{15}{m^2+5m+4} = \frac{5}{m+1} - \frac{5}{m+4},$$

so in the partial sum we get cancellation of all but three terms at the beginning and end,

$$\begin{aligned}S_k &= \sum_{m=1}^{k} \frac{15}{m^2+5m+4} \\ &= \sum_{m=1}^{k} \left(\frac{5}{m+1} - \frac{5}{m+4} \right) \\ &= \frac{5}{2} + \frac{5}{3} + \frac{5}{4} - \frac{5}{k+2} - \frac{5}{k+3} - \frac{5}{k+4}.\end{aligned}$$

So

$$\sum_{m=1}^{\infty} \frac{15}{m^2+5m+4} = \lim_{k\to\infty} S_k = 5(1/2+1/3+1/4) = 65/12,$$

by applying the algebra of limits theorem to the limit of S_k.

3 partial fraction expansion
2 for cancellation in S_k
2 for finish

End of Section B

End OF QUESTIONS